

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), MiniBatch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

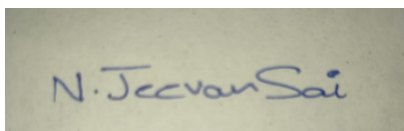
Marks: 25

Code of Conduct:

I N.Jeevan Sai (192525169) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a piece of paper with the handwritten signature "N. Jeevan Sai" in blue ink.

Name of the student:

N.Jeevan Sai

QUESTIONS:05

Total Marks:25

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

Answer:

A **high initial learning rate** allows the algorithm to make large updates, helping it reach the minimum faster. As training continues, **learning rate decay** reduces the step size, allowing fine adjustments and stable convergence.

This strategy performs better than a constant learning rate when:

- The loss function is smooth and convex.
- The initial point is far from the optimum.
- The decay schedule is chosen properly.

Advantages:

- Faster initial learning.
- Better convergence near the minimum.
- Reduces training time.

Risks:

- Too high a learning rate may overshoot the minimum.
- It may cause oscillations or divergence.
- Very fast decay may stop learning too early.

Conclusion:

A high initial learning rate with gradual decay provides both **speed and stability**, but only when the learning rate is carefully controlled.

2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2

Question:

Derive the Maximum Likelihood Estimator for μ , and analytically explain how the estimate changes when:

- The dataset contains outliers
 - The sample size increases
- What does this imply about the robustness of MLE?

Answer:

For a Gaussian distribution with known variance (σ^2), the likelihood is:

$$L(\mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

Taking the log-likelihood and differentiating:

$$\frac{d}{d\mu} \log L(\mu) = -\frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

Setting it equal to zero:



This is the **Maximum Likelihood Estimator (MLE)** for the mean.

Effect of Outliers

- Outliers significantly change the sample mean.
- The estimated mean shifts toward extreme values.
- Therefore, MLE is **not robust** to outliers.

Effect of Increasing Sample Size

- The estimate becomes more accurate.
- Variance of the estimator decreases.

- The estimate converges to the true mean (consistency).

Conclusion

MLE is **efficient and accurate** for clean data but **sensitive to outliers**, making it less robust in noisy datasets.

3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

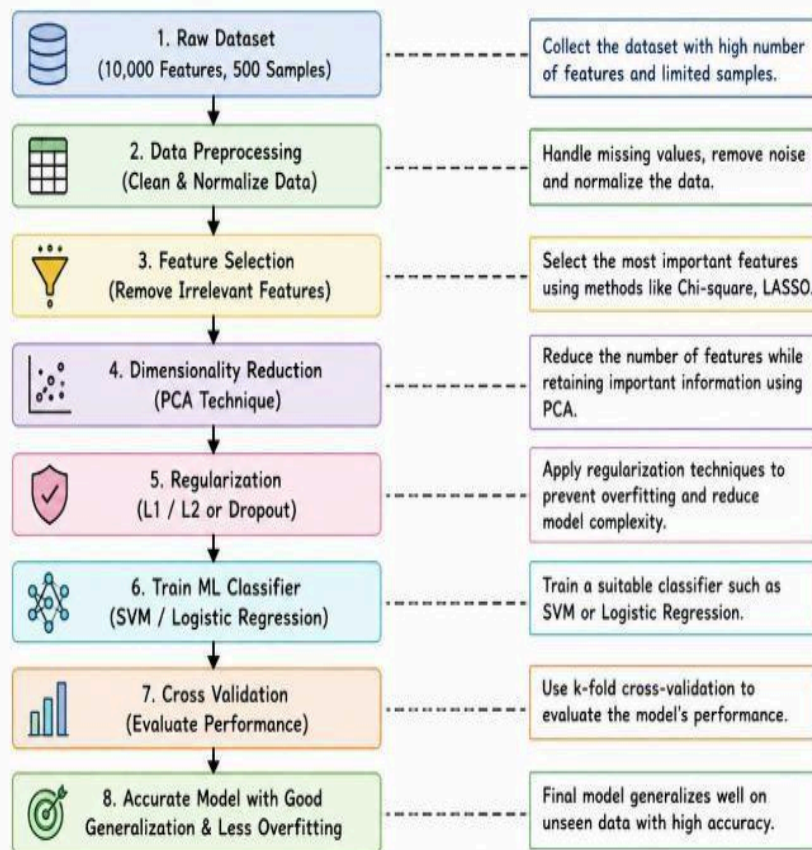
Answer:

With **10,000 features** and only **500 training samples**, the model suffers from the **high-dimensional data problem**, increasing the risk of overfitting.

Challenges

- Too many irrelevant features.
- High computational cost.
- Poor model generalization.
- Increased overfitting.

Machine Learning Pipeline for High-Dimensional Data



Recommended Pipeline

- 1. Data Preprocessing** ○ Remove missing values.
○ Normalize features.
- 2. Feature Selection** ○ Select the most important features using Chi-square or LASSO.
- 3. Dimensionality Reduction** ○ Apply PCA to reduce feature dimensions.
- 4. Regularization** ○ Use L1 or L2 regularization to prevent large weights.
- 5. Model Training** ○ Train a Support Vector Machine (SVM) or Logistic Regression.
- 6. Cross-Validation** ○ Evaluate performance using k-fold cross-validation.

Why It Works

- Reduces irrelevant features.
- Lowers model complexity.
- Improves generalization.
- Prevents overfitting while maintaining accuracy.

4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

Answer:

In an MLP, hidden neurons learn different patterns from the input data.

Adding more hidden neurons:

- Increases the network's ability to learn complex relationships.
- Allows approximation of highly non-linear functions.
- Improves representational power.

However, **more neurons do not always improve performance.**

Reasons

- The network may memorize the training data (**overfitting**).
- Model complexity increases.
- Training time becomes longer.
- Generalization on unseen data decreases.

Improving Generalization

- Use dropout.
- Apply regularization.
- Use early stopping.
- Increase training data.

Conclusion

More hidden neurons increase learning capacity, but the model must balance **complexity and generalization** to achieve the best performance.

5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Answer:

In very high-dimensional data, the **curse of dimensionality** makes learning difficult because data becomes sparse and optimization becomes more complex.

Impact on Backpropagation

- Gradients become unstable.
- Some gradients vanish or explode.
- Weight updates become less effective.

Impact on SGD

- Slower convergence.
- More noisy gradient updates.
- Requires more iterations to reach the optimum.

Impact on Generalization

- Higher chance of overfitting.
- Poor prediction on unseen data.
- Increased model complexity.

Techniques to Overcome These Problems

1. Dimensionality Reduction (PCA/Feature Selection)

- Removes redundant features.
- Simplifies optimization.
- Speeds up training.

2. Regularization (L1/L2, Dropout)

- Prevents overfitting.
- Improves model generalization.
- Controls model complexity.

Additional Methods

- Batch Normalization for stable gradients.
- Adam optimizer for faster convergence.
- Early stopping to avoid overfitting.

Conclusion

Reducing dimensionality and applying regularization improve gradient updates, accelerate SGD convergence, and produce models that generalize better on unseen data.